

MEMESCOPE

System Architecture & Default Trading Strategy

MEMESCOPE is an **opportunity intelligence platform** for the Solana meme-coin market. A scanner answers “**what exists**”. MEMESCOPE answers “**what became interesting, and why now**” — then commits that answer to a permanent, immutable record before the outcome is known.

THE PROBLEM

Thousands of tokens launch daily. Almost all are worthless. Claims about which ones mattered are made **after** the fact, when they cannot be checked.

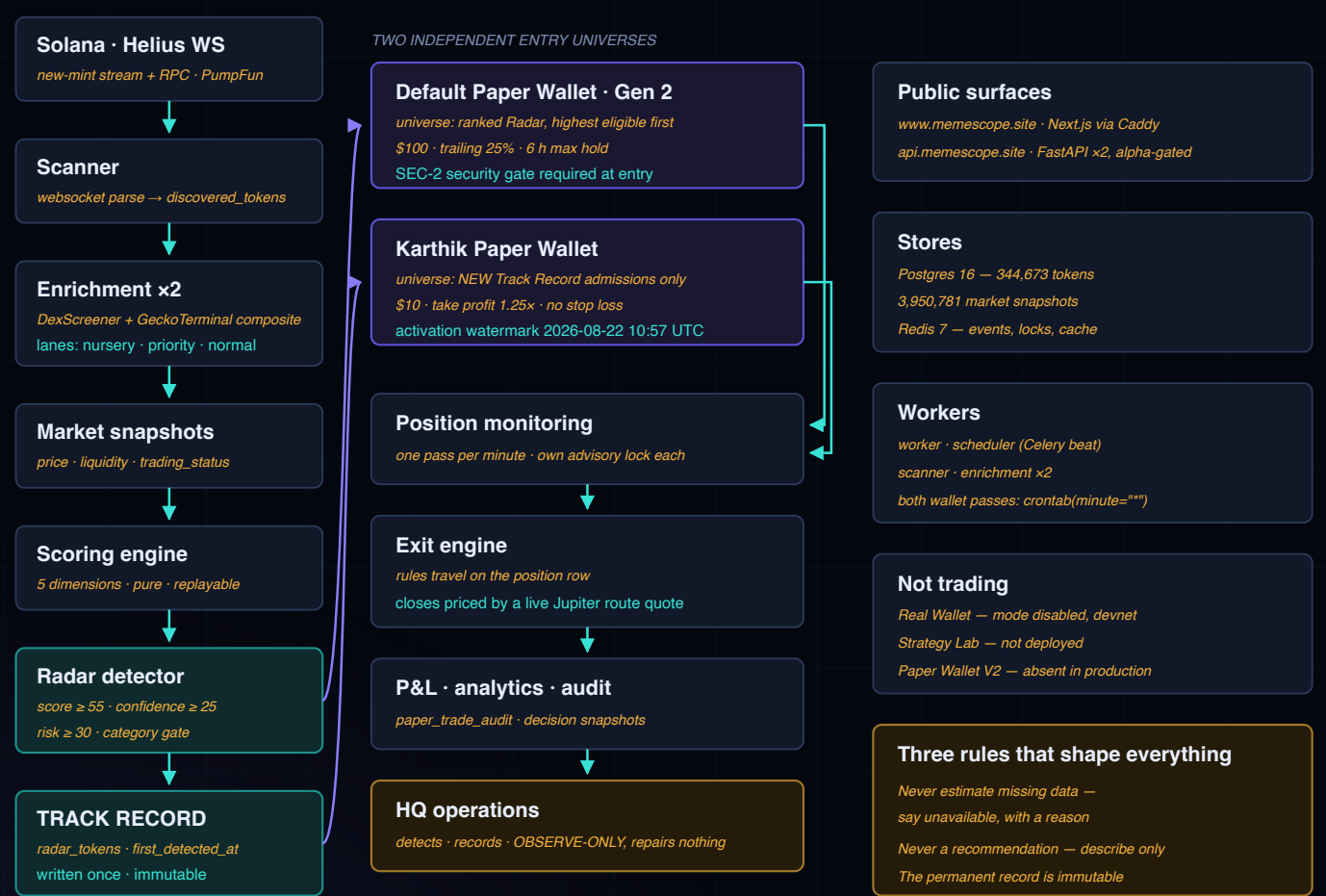
THE METHOD

Every judgement is written down **before** the outcome, with a stated rule, a timestamp, and the evidence it was based on. Missing data is charged, never estimated.

THE PROOF

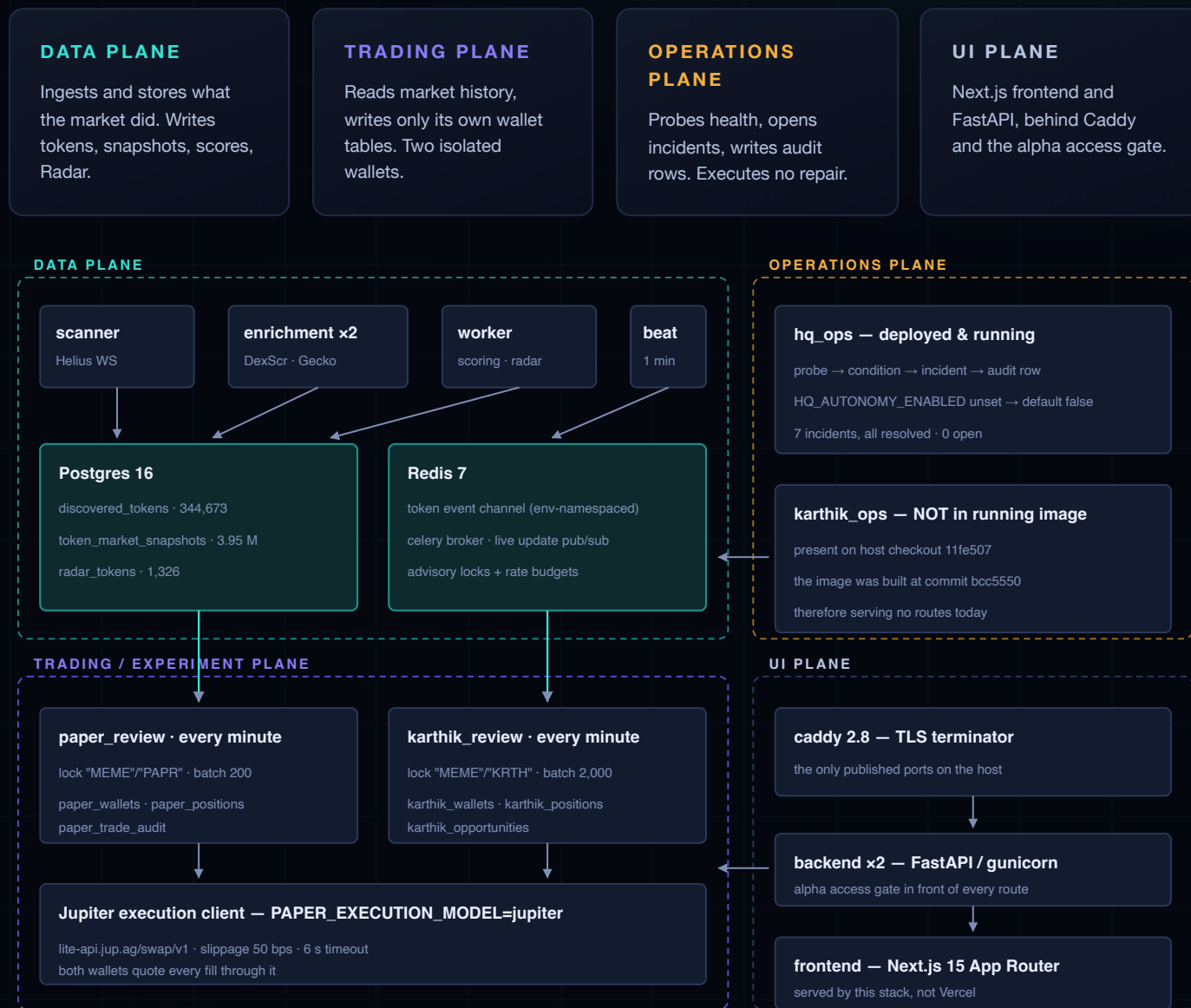
Paper wallets trade those judgements mechanically with real execution quotes. An empty board is a truthful board. Failed calls stay visible.

01 End-to-end production flow



02 The four planes

Services are grouped by what they are allowed to do. The boundary matters: the **data plane** never decides a trade, the **trading plane** never writes market history, and the **operations plane** may read everything and, today, change nothing.



Enrichment lanes — nursery: fresh tokens, ≤ 1,000, every 60 s · priority: ≤ 200, every 15 s · normal: everything else, batch 120, concurrency 4

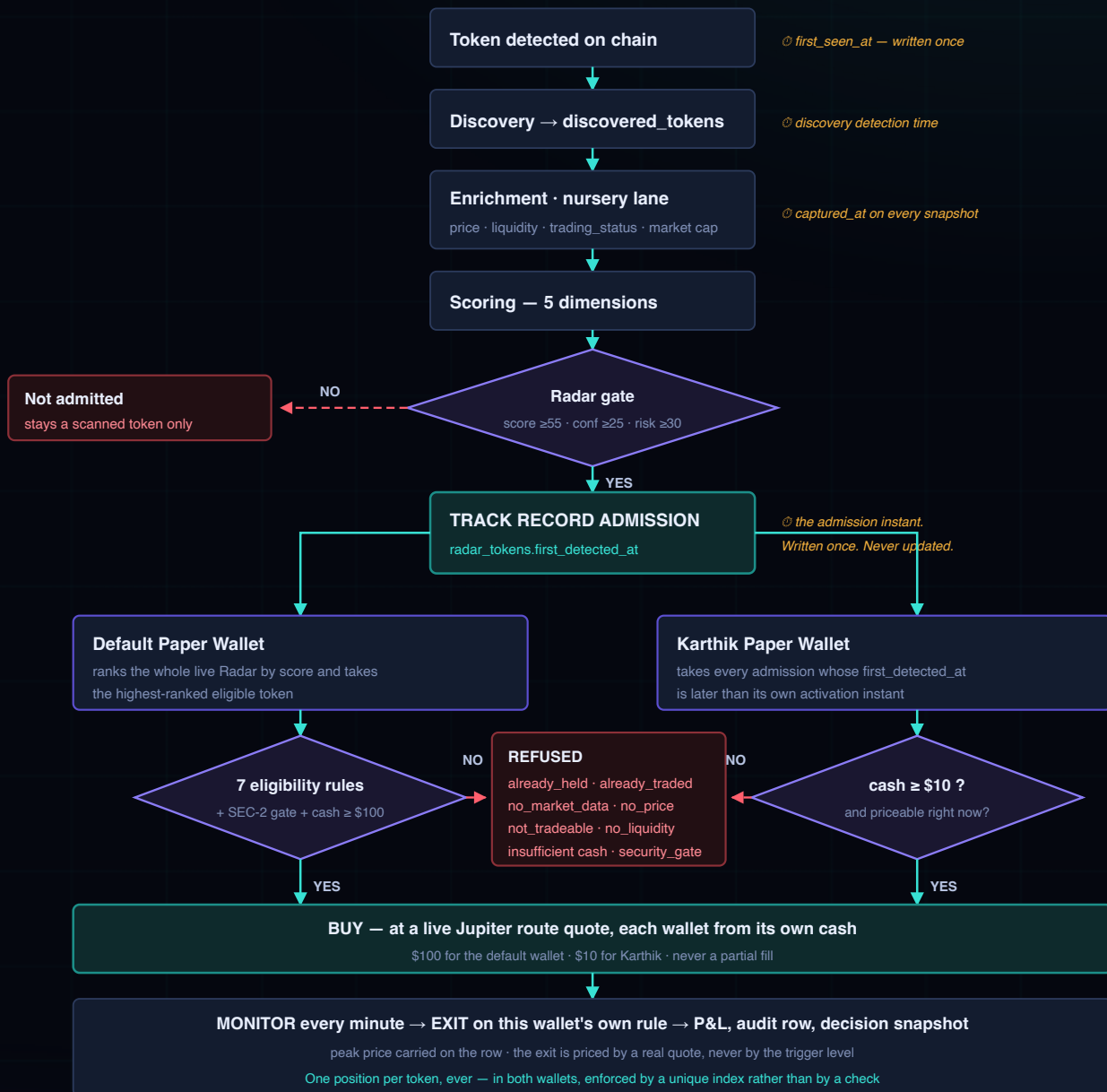
Snapshot freshness at verification: the newest market snapshot was 30 seconds old. Alembic head 0044_karthik_wallet — single head, no divergence.

Caddy is the only ingress. Nothing else on the host publishes a port; the backend replicas are reachable only over the compose network.

Correction to the assumed flow. There is no standalone “Nursery” pipeline stage between Discovery and Radar. The nursery is a **priority lane inside enrichment** that polls newly discovered tokens more often. Likewise, security (SEC-2) is not a pipeline stage before Track Record — it is an **entry precondition applied inside the wallet**, at the moment a position would be opened.

03 One token through the system

Every branch below exists in production. The rejection paths are drawn because they are the common case: of 344,673 discovered tokens, **1,326** have ever reached the Track Record — roughly four in every thousand.



TIMESTAMPS CAPTURED

first_seen_at, snapshot
captured_at,
radar_tokens.first_detected_at
(admission), position opened_at,
closed_at, last_evaluated_at,
last_market_check_at.

DETECTION VS ADMISSION

Deliberately different. A token first seen at 10:00 that was admitted at 10:20 is a **10:20 event** for Karthik. Karthik stores **detected_at** too — as **NULL** when unknown, never borrowed from the entry.

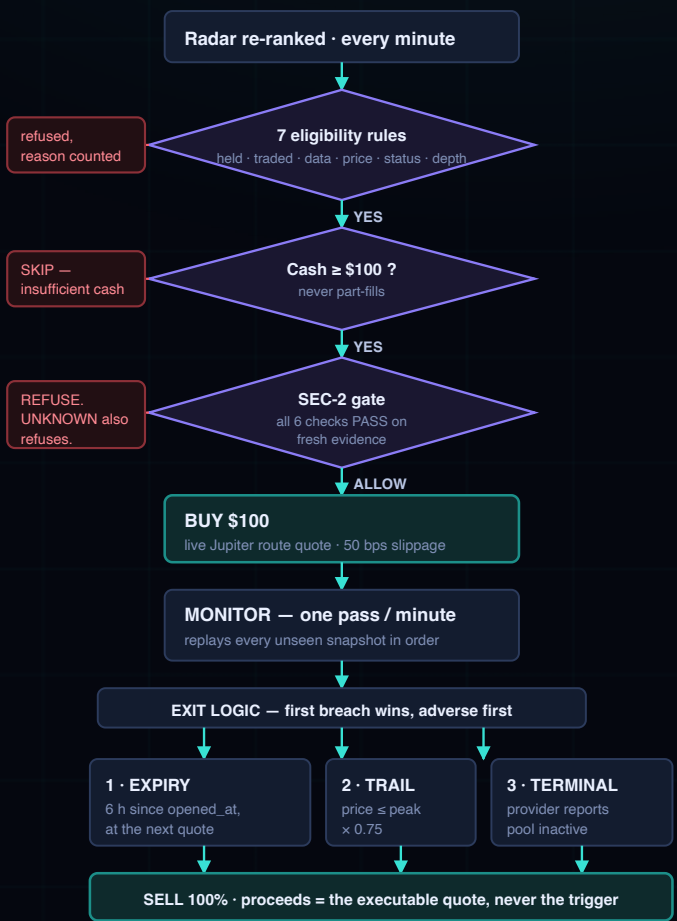
MEASURED ENTRY DELAY

Karthik: mean **29.4 s** from Track Record admission to position open, range **0.7 s – 119.6 s** across 131 entries. Bounded by the one-minute beat.

04 The default paper wallet, as production runs it

Read this first — the wallet row and the trading rules disagree. `paper_wallets` still records this wallet as generation 2, `trailing_stop_25_v1 1.0.0`. That strategy is marked **retired** in code, so the service falls back to the one operational strategy and actually trades `trailing_stop_25_secured_hold6h_v3 3.0.0-hold6h` — which is what every decision snapshot for the last 48 hours records. The rules below are the ones **being applied**. See page 7 and the limitations note for the consequence.

IDENTITY	PRODUCTION VALUE
Wallet ID	a2888cec-7624-47a5-b5bb-987762bdc459
Generation	2 LIVE
Row strategy	trailing_stop_25_v1 · 1.0.0 STALE
Applied strategy	trailing_stop_25_secured_hold6h_v3 · 3.0.0-hold6h
Started	2026-08-05 06:34:19.840157Z
Starting capital	\$1,000.00
Trade size	\$100.00 — equal weight, never partial
RULE	APPLIED VALUE
Universe	The whole ranked Radar (<code>top_n = None</code>), scanned by score
Entry	Highest-ranked eligible token, whenever cash allows
Re-entry	Never. One position per token, for the wallet's lifetime
Take profit · fixed stop	Neither . This strategy has no target and no fixed stop
Trailing stop	–25% from the highest price observed while open
Trailing reference	The high before the current reading — one snapshot cannot both set a high and fall from it
Maximum hold	6 hours , written as an absolute <code>expires_at</code> at entry
Order of rules	Adverse first: expiry → fixed stop → trailing → target
Fill	Live Jupiter route quote when the rule fires; never assumed at the trigger
Discretion	None. No rule is applied by hand



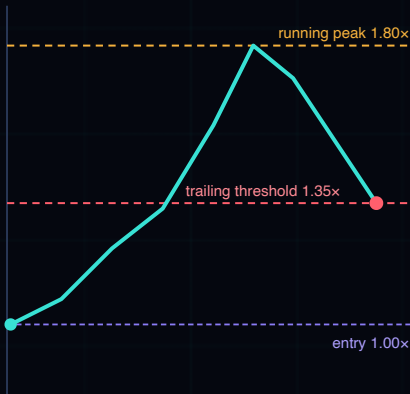
OPEN POSITIONS 2 \$200 allocated	CLOSED POSITIONS 240 stop 181 · manual 48 · expiry 11	CUMULATIVE ENTRIES \$24,200 \$1,000 recycled since 2026-08-05	DOMINANT REFUSAL cash 893 of 1,124 refusals in 24 h
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SEC-2 mandatory checks, all six must positively PASS: `MINT_AUTHORITY` · `FREEZE_AUTHORITY` · `TOKEN_PROGRAM` · `TOKEN_EXTENSIONS` · `VENUE` · `LIQUIDITY_SECURITY`. **UNKNOWN blocks the buy without labelling the token unsafe** — an RPC outage is not evidence of a dangerous token, but it is not good enough to buy against either. Only `TOKEN_EXTENSIONS` may answer `NOT_APPLICABLE`.

05 What \$100 actually does

Three price paths, one rule set. The **running peak** is stored on the position row and only ever moves up; the **trailing threshold** is always 75% of it. The position closes at the first breach, and the money it returns is what a route would actually pay — not what the trigger said.

A · WINNER — the trailing stop does its job

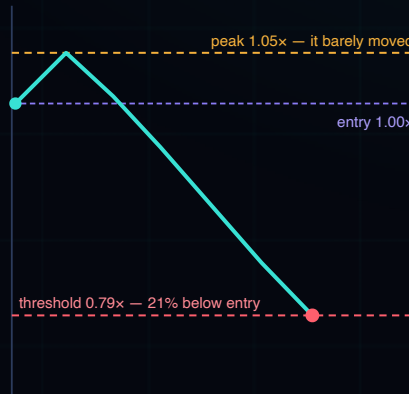


BUY \$100 → SELL ≈ \$135

+\$35 gross

Gave back 25% of a high well above cost.

B · ORDINARY LOSS — it never went up

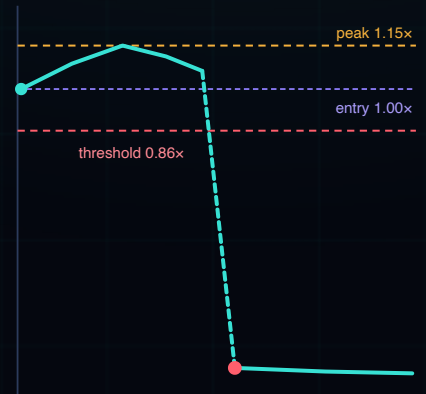


BUY \$100 → SELL ≈ \$79

-\$21

A position that never rises still gives back 25% of its own entry.

C · GAP / RUG — the rule fires too late



BUY \$100 → SELL at what a route pays ≈ \$6

-\$94, not -\$14

One snapshot later the pool is drained. The threshold was skipped over.

WHY A 25% TRAILING STOP DOES NOT CAP THE LOSS AT 25%

1 · A stop is a trigger, not a fill. The rule says **when** to sell. The market says **at what**. Production deliberately books the executable quote, never the trigger level — so if the price gapped straight through the threshold, the fill is below it by however far it gapped.

2 · The market is sampled, not watched. Positions are evaluated once a minute against snapshots that are themselves polled. Between two readings a meme-coin pool can be drained entirely. **Nothing traded at the threshold, so nothing could have sold there.**

3 · The threshold is 25% off the peak, not off cost. A position that ran to 1.80x and gave back 25% is a **+35% winner**. A position that never rose gives back 25% **of the entry** — and if the pool dies first, the provider marks it terminal and it returns **\$0.00**.

Evidence, not assertion: the sibling Karthik wallet — same execution path, same market, exit target instead of trailing stop — has closed **27 positions at exactly \$0.00**, on the provider's own report that the pool had nothing left to sell into. That outcome is available to this wallet on identical terms.

ALSO: THE 6-HOUR CUT

Since 2026-08-21 every new position carries an absolute **expires_at**. A position still climbing at hour six is sold anyway, at whatever is executable. **11 of 240** closed trades exited this way.

COSTS ARE CHARGED, NOT ASSUMED AWAY

Fills come from live Jupiter routes (50 bps slippage tolerance). Where a route is unavailable the fallback model charges **30 bps per side** plus exact constant-product impact $S^2 / (L/2)$.

WHAT THE NUMBERS ABOVE ARE

Illustrative price paths on the **real, verified rules** — chosen to show each branch. They are **not** replays of specific production trades. Production outcomes are on page 4 and page 7.

06 How this loses money, and how it breaks

Nothing here is hypothetical framing. Each item names the mechanism, what production does about it today, and what remains exposed.

RUG / GAP-THROUGH

Liquidity is pulled between two snapshots; the trailing threshold is never traded.

Mitigated: the fill is the executable quote, so the loss is booked honestly. **Exposed:** the loss itself is uncapped — up to the full \$100.

LIQUIDITY COLLAPSE

The pool thins until no route will quote the position at all.

Mitigated: a pool reported **inactive** closes at \$0.00, not at its last printed price. **Exposed:** that is a real, total loss — not a bookkeeping artefact.

STALE MARKET DATA

A position judged against a reading older than the market it claims to describe.

Mitigated: exits replay **every** unseen snapshot in timestamp order, and expiry is measured against each observation's own timestamp — never a wall clock. A worker that missed six hours produces the same trades as one that missed none.

FEED OUTAGE

Enrichment or a provider stops; positions stop being re-priced.

Mitigated: positions are **held and said to be held**, never closed on absent data; HQ opens an incident. **Exposed:** during the gap no exit rule can fire, whatever the market does.

INSUFFICIENT CASH · CAPITAL SATURATION

Qualified tokens arrive with no money left to buy them.

The dominant condition today: 893 of 1,124 refusals in 24 h were **insufficient_paper_cash**. With no take-profit and no fixed stop, capital sits in open positions until a trailing stop or the 6-hour clock releases it.

EXECUTION PRICE VS TRIGGER PRICE

The price that fires a rule is not the price that fills it.

Production stores both — **exit_observed_price** and **exit_price**. On the sibling wallet, average **entry price impact** was 10.24%, exit impact 0.87%. An order against a thin pool moves the price it is trying to get.

SEC-2 LIMITATION

The gate proves specific, checkable properties. It is not a safety verdict.

It **cannot** detect a coordinated exit by holders, a social rug, or a team that simply stops. LP holders are not resolved — an outstanding LP supply is UNKNOWN, which refuses the buy without calling the token unsafe.

YOUNG-TOKEN RISK

The Radar has no minimum age and, by design, no market-cap floor.

Admission claims something **changed**, not that a project is durable. Bonding-curve pairs report no liquidity, so they are refused rather than costed against an assumed depth — correct, and it removes a whole class of otherwise-eligible tokens.

TWO FAILURE MODES SPECIFIC TO THIS DEPLOYMENT, FOUND DURING VERIFICATION

FINDING

The wallet row records **trailing_stop_25_v1**, but the wallet trades **trailing_stop_25_secured_hold6h_v3**.

PAPER_WALLET_STRATEGY_ID is set to a retired strategy id.

CONSEQUENCE

The repository's SEC-2 backstop keys off the **row**, and **trailing_stop_25_v1** is not in the gated set — so the last-line defence is **inert for the live wallet**. The service-level gate is working (27 **security_gate** refusals in 24 h), but the defence-in-depth check built to catch a future code path that skipped it is not armed.

The service logs **paper_strategy_unknown** on every pass and falls back to the registry default. The wallet trades correctly by accident of a safe fallback, not because it was configured to.

07 Seven systems — status as deployed

SYSTEM	STATUS	WHAT IT IS, IN PRODUCTION, TODAY
Track Record	LIVE	The permanent record. <code>radar_tokens</code> , 1,326 rows, <code>first_detected_at</code> written once. Feeds Karthik's universe directly.
Default Paper Wallet	LIVE	Generation 2 · own \$1,000 · \$100 per trade · own tables. Enters from the ranked Radar . 2 open, 240 closed.
Karthik Paper Wallet	LIVE	Own \$1,000 · \$10 per trade · separate tables , separate advisory lock, separate Celery task. Enters from Track Record admissions . Activated 2026-08-22 10:57 UTC.
HQ Operations	OBSERVE-ONLY	Deployed and running. Probes, opens incidents, writes audit rows. <code>HQ_AUTONOMY_ENABLED</code> unset → defaults false , so it executes no repair . 7 incidents, all resolved.
Agent Karthik <code>karthik_ops</code>	NOT DEPLOYED	On the host checkout (<code>11fe507</code>) but absent from the running image , built at <code>bcc5550</code> . Prod compose uses <code>volumes: !reset []</code> — no bind mount — so disk never reaches the process without a rebuild. Serves no routes today .
Strategy Lab	RESEARCH	No module and no tables exist in production. Built and verified locally only.
Paper Wallet V2	ABSENT	<code>PAPER_V2_MODE</code> exists in neither the runtime env, the image, nor the host checkout. “V2” is the historical name of the Sprint 30 relaunch that created generation 2 — it is the default wallet.
Real Wallet	EXEC LOCKED	<code>MODE=disabled</code> , <code>EXECUTION_ENABLED=false</code> , <code>AUTOTRADE=false</code> , dry-run false, network devnet , entry size \$0 , no public key. 0 positions . Nothing has ever executed .



VERIFIED AGAINST PRODUCTION

ENVIRONMENT	production
HOST	vps-651730bb · 51.79.166.133 · OVH
PUBLIC WEBSITE	www.memescape.site HTTP 200
PUBLIC API	api.memescape.site reachable · alpha-gated
SERVING COMMIT	bcc5550c2baf07aea484ae97e64de2f064626031 image built 2026-08-22T10:39:44Z · up 10:45:49Z
HOST CHECKOUT	11fe5077aef76e086166a70ba9007b55c60f6401 NOT built · NOT serving
ALEMBIC HEAD	0044_karthik_wallet — single head
STRATEGY ID	applied: trailing_stop_25_secured_hold6h_v3 wallet row: trailing_stop_25_v1
STRATEGY VERSION	applied: 3.0.0-hold6h · wallet row: 1.0.0
GENERATED	2026-08-22 14:08 UTC

ASSUMPTIONS & LIMITATIONS

- Wallet figures are a **snapshot at 2026-08-22 14:08 UTC**. Both wallets trade every minute; the numbers have moved since.
- Page 5's price paths **illustrate the verified rules**; they are not replays of specific production trades.
- Refusal counts are the trailing 24 h of `paper_decision_snapshots`. Execution-impact figures are averaged over Karthik's 131 entries, which share this wallet's execution client but not its \$100 order size.
- The API could not be queried programmatically — the alpha gate rejects scripted sign-in. Figures come from production **source, container environment and database**, which are the facts the API renders.
- Production moved from `bcc5550` to `11fe507` **on disk** during verification. Every statement here describes the **running image**, still `bcc5550`.
- No production data, configuration, strategy or wallet was altered to produce this document.